

ANIMA

A Developmental Approach to a Self-Organizing Forager

Technical white paper · Research record · Reproducible

ANIMA: A Developmental Approach to a Self-Organizing Forager

Experimental Findings, Registered Negative Results, and the Two Walls

Status: White paper · Research record · September 2026 System: world_survival (crates/anima-world), ANIMA developmental artificial nervous system in Rust Reproducibility: deterministic (seeded), byte-identical defaults, all mechanisms env-gated

1. Executive Summary

ANIMA is a from-scratch, self-organizing artificial nervous system: a spiking LIF network receives retinal/proprioceptive frames from a physical 3D body in an arena and controls it to forage for food. Under a binding charter it uses no hand-designed internal architecture, no pretrained models, no external memory modules, and the only reward is food.

This paper records a full characterization campaign. We found and fixed a real control bug, built a working forager, gave it a second drive (a home/rest site), and confirmed it can produce repeated return-home behavior when that is rewarded. But we also isolated — with measured evidence, not inference — two walls that block the original open-ended goal:

1. Internal-band neural saturation: the network's internal (expansion) layer fires at 100% every tick (out_sum 6000, kc_spikers 256/256). Every reward-gated learning mechanism therefore becomes a non-selective global wash that the evolutionary loop ignores. 2. Fitness-economy ceiling: evolution pins at ~121-137.6 fitness regardless of mechanism stack.

A four-lever campaign (competition, spike threshold, input amplitude, per-neuron heterogeneity) failed to de-saturate the band — and revealed it is temporally bistable: cold→sparse, then a sharp phase-transition to hot→saturated. Selective learning requires sustained sparsity and is therefore structurally absent. Tracked next levers are listed in §9.

2. Motivation

Goal: an organism that autonomously forages — finds food, consumes it — and, ultimately, self-improves open-endedly ("make it like us"). The project favors laboratory discipline over best-effort results: every mechanism is registered, falsifiable, and reproducible.

3. The Agent and the World

- Body: 6-DOF physics (yaw/pitch/thrust/strafe/lift/brake), arena 100×100, gravity 9.81 applied in the simulator only, ground clamp, energy tank (starts 100, drain 1.0/beat, death at 0).
- Sensory: 48-channel retina (5 primitives, analytic ray raster) + 8 proprioceptive channels, plus optional 6-channel food-smell (body-relative gradient).
- Network: LIF spiking, 1 ms ticks, seeded Xoshiro256PlusPlus, STDP + structural growth (V2), 24-56 inputs / 40-(128)-(256) internal / 12 motor outputs.

- Reward: food only. Steering is the action, never rewarded directly.
- World dynamics: food placement (static seed / curriculum / escalating / stable), energy refill on contact, death at tank zero.

4. Binding Constraints (the Charter)

- No hand-designed internal architecture; no pretrained/CNN/CLIP; no external memory modules.
- All mechanisms env-gated; with every gate unset, output is byte-identical to a frozen baseline.
- Determinism: same args + env $\Rightarrow$ identical output (registered).
- Pre-registration: expected direction and success bars are set before running; negatives are recorded, constants are not tuned on outcomes.
- Charter amendments (e.g. adding a home need) require explicit operator approval.

5. Method

Protocol. Each mechanism is introduced env-gated, then verified: (1) default identity, (2) determinism (two-run diff), (3) observability (gate on vs off), (4) workspace suite, (5) a registered success bar. Failures are recorded as evidence-backed negatives — never silently retuned. Results below are measured; unverified causal claims are flagged as such.

6. Measured Results

6.1 Verified wins

#	Mechanism	Result	Evidence
1	Control bug fix	Steering yaw channel was sign-inverted; body flew away from food	post-fix food_dist 11.5 $\rightarrow$ 6.7, touches 0 $\rightarrow$ 9 at reachable radius
2	Foraging under shaping	Up to 126 touches / life; strong local pursuit	curriculum + smell stack
3	SL2_STASIS (stable food + re-entry guard + tank cap)	3 $\rightarrow$ 36 genuine touches / life; energy bounded; no farm	unsaturable only via geometry-exact re-entry guard
4	SL2_HMEXP (hunger-modulated search)	settle-when-fed (0.4 $\times$ noise floor), roam-when-hungry; early-survivor signal in	single-life $\approx$ baseline; evolve best 0 $\rightarrow$ 33.9 by gen 5 (8-gen window)
5	SL2_NEST + SL2_NESTFIT (home/rest second drive + homing)	First repeated return-home behavior (elite ret 1 $\rightarrow$ 3 on seed 7)	home = nearest non-food landmark; stamina = capability constraint; reward per return
6	Proactive survival (SL2_PROG)	survived 2230 beats (vs $\sim$ 60–200), food escalates until defeated	big lifespan + escalating difficulty

6.2 Registered negative results

Mechanism	Outcome	Notes
SL2_NOVELTY (MCC archive)	observably steers selection; ceiling unchanged	rank-order changes, no escape
SL2_RSTDPP (dopamine $\times$ STDP)	fires correctly (4.9 $\times$ Δw in window) but invisible to	$\sim$ 1e-3/synapse below mutation floor
SL2_MARS / SL2_DENSITY	environmentally observable; ceiling unchanged	same survivor, bigger number
SL2_MB (mushroom-body locus)	wiring verified firing; no sparse premise	256/256 KCs eligible — non-selective
De-saturation campaign (§7)	all four levers bounded negative	competition / threshold / amplitude / heterogeneity

Cross-cutting finding: within-life plasticity (any locus) is $\sim$ 1e-3–0.24/synapse but the GA's selection is death-gated and coarse ($\pm$ 10% mutation) — so learning that is not selectivity-visible never changes outcomes. Evolution "listens" only to selection-level changes (novelty rank-order, the homing reward NESTFIT — the one lever that moved behavior).

7. The Two Walls

7.1 Wall 1 — Internal-band saturation $\Rightarrow$ selective learning is impossible

The internal/expansion layer fires at 100% every tick. Under this regime any reward-gated plasticity is a uniform global wash the GA trivially mutates around (measured: evolve trajectories byte-identical with and without the reward bump).

De-saturation campaign (all measured):

Lever	Range	Result	Mechanism
Pairwise competition (SL2_INH/SL2_INH)	0.5-4.0	256/256 always	Poisson drive rebuilds the one-tick inhibitory dip
Spike threshold (SL2_THETA)	2-8×	256/256 always	52-amplitude input drive crosses any spread
Input amplitude (SL2_AMPL)	52→3	100%→0% cliff	uniform scaling is bistable — no graded regime
Heterogeneity (SL2_HET)	0.2-1.0	256/256 alone; transient at corner	lognormal threshold spread can't sustain

Final mechanism (discriminator, KCPROBE_EVERY=10): the band is temporally bistable — cold at life start (2/256), then a sharp phase-transition to saturated by beat ~25-30. It is a collective recurrent-excitation/hysteresis flip, not a gradual slow-state ramp (a linear rate-constant knob likely won't tame it).

7.2 Wall 2 — Fitness-economy ceiling

Evolution converges to a scoring-family-dependent pin (not one flat number):

- Plain-survival scoring (final_energy + 3·touches): pins 121.8 / 121.0 (HMEXP+NEST, seeds 1/2).
- Density/ARS scoring (adds 20·touches²/beats and 4·persistence): pins 137.5-137.6

(ARS run seed 7; the higher number is the bonus terms, not a real escape).

- NESTFIT scoring (plain + 40·nest_returns): 121.8 / 179.4 (seed 1 / seed 7), where 179.4

includes +40×3 from three returns — again additive, not a genuine survival gain.

final_energy + 3·touches saturates once food is reachable; across every scoring family the ceiling is the fitness-economy saturation, and the ~150-180 bests are additive bonuses.

8. Discussion

- The organism is real and capable: it forages strongly, survives long, and can sustain a home-return behavior when rewarded — the first causal win for the "second drive" direction.
- The walls are structural, not tuning: saturation is immune to competition/threshold/amplitude/heterogeneity at every tested magnitude, and the fitness economy is a property of the reward shape.
- The iron rule of the campaign: within-life learning cannot clear a coarse, death-gated selection; only selection-level changes are evolution-visible.
- Full-stack non-additivity: running the coherent verified organism

(STASIS+HMEXP+NEST+NESTFIT+OINH+pool128) reproduces the NESTFIT baseline exactly at 15 gens (seed 20260924: 121.8/ret=0; seed 7: 155.0/ret=1) — the mechanisms are orthogonal to the fitness economy, not additive.

8.5 Limitations

The claims above are bounded by the measurement window and should be read with these limits:

- Small N seeds: only two evolve seeds per run; seed-dependence (e.g. elite_ret 0 vs 3) is a real finding but a 2-seed sample cannot establish robustness.
- Short generation windows: several series (MB, full stack) ran 12-15 gens per seed within a 3500 s tool cap; pin values at longer horizons (e.g. 179.4/ret=3 at gen 20-30) come from a subset of runs.
- Single-threaded CPU stack: 256-pool evolution exceeded the hour cap; some negatives (MB locus, larger expansions) were bounded by compute, not measured to conclusion.

- Partial windows: a few registered negatives rest on overlapping-generation comparisons (e.g. gens 0-5) rather than full-architecture sweeps.
- No GPU, no parallelization; determinism is enforced serially.

9. Recommended Next Steps (registered levers)

1. Attack the phase-transition, or keep the band cold: target the recurrent-excitation/hysteresis that flips the band to saturation, or maintain continuous-cold sparsity — the precondition for any selective-learning re-test (SL2_MB locus remains unproven, not falsified). 2. Reform the fitness economy: sustained-foraging / survival density shaping so selection has a gradient to climb past the ceiling. 3. GPU/parallel step (engineering): the fly-connectome literature runs 138k neurons real-time on GPU; this single-thread stack can't afford a biological-size expansion in evolution. Removing the compute tax unblocks every experiment above.

10. Reproducibility

- Deterministic by construction (seeded LCG-free Xoshiro; no HashMap iteration order).
- Frozen default identity: world_survival 20260924 77 60 ⇒
WL2 RESULT beats=60 died_at=none final_energy=40.0 food_touches=0 novel_flags=0.
- Test rig: KCPROBE/SW_PROBE/MB_PROBE/NEST_PROBE (env-gated, stderr) + registered
SL2_WL2 knobs; workspace suite 198/0.
- Workflow discipline captured in the anima-experiment-protocol managed skill.

Appendix A — Reproduction specification

Everything needed to recreate the experiments, with exact equations and registered constants. All mechanisms are env-gated; with no env set, the default run and its frozen output are byte-identical to the recorded baseline.

A.1 World & body dynamics

- Start energy 100.0; SL2_PROG overrides to PROG_LIFE 4000.0 (life-span mode).
- Per beat: energy -= 1.0 (plus explore metabolic cost METABOLIC_SPEED × |vel|); on food contact within 6.0 units, energy += 30.0; death exactly at energy == 0.
- SL2_STASIS (stable food): food does not teleport on touch. Re-entry guard — a touch counts

only if the body left the 6.0 eat radius since the last counted touch; tank caps at the life-start energy (min(energy, tank_cap); tank_cap re-read after any PROG override).

A.2 Network construction (build_net)

Defaults: connectivity 0.038, w_init 0.2, amplitude 52.0, adaptation_tau_ms 200.0, adaptation_gain 0.05, inhibition_gain 0.0, slow_state_tau_ms 5000.0, slow_state_beta 0.0046875, latch_enable true, theta_rel_mean 1.0, theta_rel_sd 0, u_plateau_rel_sd 0, tau_het_rel_sd 0, phi_rel 0.5, eta_rel 0.0, output_inhibition_gain 0.0, d58_reflex K (K = the 8-neuron reflex/familiarity band that follows the outputs; see const K: usize = 8). Env overrides: WL2_OINH → output_inhibition_gain 0.15; SL2_POOL → internal pool size (default 40); SL2_MB → pool 256, connectivity 0.10; SL2_IINH → internal_inhibition_gain; SL2_THETA → theta_rel_mean; SL2_AMPL → amplitude; SL2_HET → theta/u_plateau/tau_het_sd (per-neuron lognormal, applied after wiring). Wiring is seeded-via-Xoshiro with no HashMap iteration order, so identical args+env ⇒ identical network.

A.3 Fitness per scoring family (evolution evolve_run)

D-38 death gate: a dead organism scores 0 unconditionally. Alive:

- Plain-survival: $\text{fit} = \text{final_energy} + 3 \cdot \text{touches}$ (TOUCH_BONUS = 3.0)

- Density (SL2_DENSITY): $+ 20 \cdot \text{touches}^2 / \text{beats}$

- ARS (SL2_ARS, subsumes density): $+ 4 \cdot \text{persistence}$, where

$\text{persistence} = \sum T \exp(-\text{gap_k} / 20)$ over consecutive touches (beats since previous touch)

- NESTFIT (SL2_NESTFIT, adds to any alive base): $+ 40 \cdot \text{nest_returns}$

(nest_returns = count of false→true re-entries into the nest radius — see A.5)

A.4 Behavioral drives (env-gated)

- Hunger: $\text{hunger} = 1 - \text{energy} / \text{start_energy}$.

- SL2_HMEXP (hunger-modulated exploration): motor-noise gain

$\text{noise_scale} = 0.4 + 0.6 \cdot \text{hunger}$ (fed $\Rightarrow 0.4 \times$ baseline noise, hungry $\Rightarrow 1 \times$). Applied to all three explore-noise sites.

- SL2_RSTDPP (three-factor): on food touch set dopamine = 1.0; per beat dopamine $\times = 0.6$;

STDPP gain = $\min(1 + 3 \cdot \text{dopamine}, 4)$.

- SL2_NOVELTY (MCC selection): behavior characterization $\text{bc} = [\text{x_bin}, \text{z_bin}, \text{alive}]$ where

$\text{x_bin} = \text{clamp}((\text{px}+50)/10, 0.9)$, same for z; novelty of an agent = $\sum \text{archive}(\text{diff}(\text{a}, \text{bc}))/3 / \max(|\text{archive}|, 1)$; selection rank key (alive, novelty, fit) descending; archive cap 50 (evict oldest).

A.5 Home / rest (SL2_NEST + SL2_NESTFIT)

- Nest = the non-food scene primitive nearest the spawn (deterministic).

- Stamina (capability constraint): starts 100; away from the nest,

$\text{stamina} -= 0.4 \times |\text{vel}|$ per beat; within NEST_RADIUS = 8.0 of the nest, $\text{stamina} += 60$ (cap 100). Motor rates scale by $\min(1, \text{stamina} / 25)$ — exhaustion immobilizes.

- nest_returns increments on each false→true entry into the 8.0 nest radius.

A.6 Mushroom-body locus (SL2_MB)

Sparse expansion (256 KC, ~6 inputs/KC) + reward-gated input plasticity at the KC band: per touch, for every KC with eligibility $e > 0$, its input synapses get $\Delta w = 0.03 \cdot e \cdot 8$ (clamp 1.0). Measured saturation (A.7) currently makes $e > 0$ for all 256 KCs, so the bump is non-selective.

A.7 De-saturation measurements (the four-lever campaign)

KCPROBE (env-gated) reports distinct internal-band spikers per sampled beat (KCPROBE EVERY sets the beat interval). Results: pairwise competition 0.5–4.0 $\Rightarrow$ 256/256; threshold 2–8 $\Rightarrow$ 256/256; amplitude 52/20/8 $\Rightarrow$ 256/256, amplitude 3 $\Rightarrow$ 0/256 (a cliff); heterogeneity 0.2–1.0 $\Rightarrow$ 256/256 alone, and at amplitude=4, het=1.0 the band is temporally bistable: beat 0 = 2/256, beat 20 = 3/256, beat 30+ = 256/256 (a sharp phase transition).

A.8 Verification battery and frozen identity

1. Build: `cargo build --release -p anima-world --bin world_survival`. 2. Default identity (no env): `./target/release/world_survival 20260924 77 60` must print exactly `WL2 RESULT beats=60 died_at=none final_energy=40.0 food_touches=0 novel_flags=0`. 3. Determinism: run any gated config twice, diff empty. 4. Suite: `cargo test --workspace` $\rightarrow$ 198 passed / 0 failed. 5. Probe output is on stderr — capture with `2>&1 >/dev/null | grep`, not `2>/dev/null`.

A.9 Provenance notes

The 36-touch (STASIS) and 2230-beat (SL2_PROG) figures are recorded-lesson values (session-learn log);

logs for those long runs were overwritten by later runs, so they carry recorded provenance rather than a surviving artifact file. Every other cited number in this paper was verified directly against a surviving run log.

A.10 Genetic algorithm structure (evolve_run)

- Population $N = 4$ per generation; seeded net = build_net(net_inputs(smell), derive_seed64(esec,0,i)).
- Per-generation curriculum ramp: gen_start = min(8 + 3·g, 40).
- Selection: deterministic sort by fitness (or (alive, novelty, fit) under SL2_NOVELTY);

death gate (dead $\Rightarrow$ fit 0) holds.

- Elite carry: order[0] carried unchanged (weights copied via breed_inner, mutate=false);

after the same seed is best for ELITE_STALL_GENS = 5, the elite is carried with mutation (Fix B, anti-stall).

- Offspring: best $\rightarrow$ 2, second $\rightarrow$ 1; breed_inner(parent, derive_seed64(esec,g+1,slot), mutate=true)

copies parent weights by (pre,post) and mutates each with $p = 0.10$, amplitude 0.10 (W_MUT_P / W_MUT_AMP), then re-adds parent-born V2 synapses while under the growth cap.

- SL2_NOTARGET suppresses the early TARGET break (best > 40); --generalize is champion mode

(probes the elite per gen on BEATS_LONG natural food, stops at ULTIMATE_BAR).

- Beats per life: BEATS_EVOLVE = 100 (evolve+explore), BEATS_LONG for champion probes.

A.11 Env-gate $\rightarrow$ exact-run-command table

Core (documentation only, not all commands run daily):

- Default / identity gate: ./target/release/world_survival 20260924 77 60
- Stability + settle + home reward (full stack):

```
SL2_STASIS=1 SL2_HMEXP=1 SL2_NEST=1 SL2_NESTFIT=1 WL2_OINH=1 SL2_POOL=128  
SL2_NOTARGET=1 ./target/release/world_survival --evolve --explore --smell --until 15
```

- Per-mechanism observability runs use the same binary with a single gate added; probes

(KCPROBE, MB_PROBE, SW_PROBE, NEST_PROBE) are env-gated and print to stderr — capture 2>&1
>/dev/null | grep <probe>.

- Seeds used: 20260924 and 7 (both EVOLVE_SEEDS).

11. Selected References

- FlyWire adult Drosophila connectome and its simulation in embodied bodies (NeuroMechFly, Nat.

Methods 2024; whole-brain embodied fly sims; Fly-connectomic graph models) — evidence that small spiking networks drive 3D behavior and that large-scale sims run real-time on GPU.

- Mushroom-body associative learning (sparse random expansion + dopamine-gated input plasticity;

trace conditioning; biased PN-KC connectivity) — the biological scheme ANIMA's SL2_MB was drawn from, whose sparse-coding premise the saturation wall currently denies.

- Novelty search / minimal-criterion coevolution (Lehman & Stanley; MCC) — the selection-level

change that observably steers but does not escape the ceiling.

All results in this paper are reproducible from the repository; every claim is grounded in a measured run recorded in the session-learn log. Unverified causal attributions are explicitly marked.